

This process represents the linux memory allocation

```
hari@LAPTOP-C63DNL02:~/OS/2520090048_Practical/Practical_7$ gcc -Wall -Wextra -g memory_layout.c -o memory_layout
compilation terminated.
hari@LAPTOP-C63DNL02:~/OS/2520090048_Practical/Practical_7$ gcc -Wall -Wextra -g code.c -o code
hari@LAPTOP-C63DNL02:~/OS/2520090048_Practical/Practical_7$ ./code

=====
Linux Process Memory Addresses
=====
PID                               : 1675

--- CODE / TEXT ---
Address of main()                 : 0x56b0261f4243
Address of code_function()       : 0x56b0261f4229

--- GLOBAL VARIABLES ---
Address of global_var            : 0x56b0261f7010
Address of uninitialized_global : 0x56b0261f701c

--- STATIC VARIABLE ---
Address of static_var            : 0x56b0261f7014

--- HEAP ---
Address returned by malloc       : 0x56b062c322a0

--- STACK ---
Address of stack_var            : 0x7ffc0d9adb5c

Keep the process running...
Press ENTER to terminate.

hari@LAPTOP-C63DNL02:~/OS/2520090048_Practical/Practical_7$
```

While this is running the process

```
hari@LAPTOP-C63DNL02:~/OS/2520090048_Practical/Practical_7$ cat /proc/12345/maps
cat: /proc/12345/maps: No such file or directory
hari@LAPTOP-C63DNL02:~/OS/2520090048_Practical/Practical_7$ cat /proc/1675/maps
56b0261f3000-56b0261f4000 r--p 00000000 08:30 229134 /home/hari/OS/2520090048_Practical/Practical_7/code
56b0261f4000-56b0261f5000 r-xp 00001000 08:30 229134 /home/hari/OS/2520090048_Practical/Practical_7/code
56b0261f5000-56b0261f6000 r--p 00002000 08:30 229134 /home/hari/OS/2520090048_Practical/Practical_7/code
56b0261f6000-56b0261f7000 r--p 00002000 08:30 229134 /home/hari/OS/2520090048_Practical/Practical_7/code
56b0261f7000-56b0261f8000 rw-p 00003000 08:30 229134 /home/hari/OS/2520090048_Practical/Practical_7/code
56b062c32000-56b062c33000 rw-p 00000000 00:00 0 [heap]
73024aa00000-73024aa28000 r--p 00000000 08:30 46312 /usr/lib/x86_64-linux-gnu/libc.so.6
73024aa28000-73024abb1000 r-xp 00028000 08:30 46312 /usr/lib/x86_64-linux-gnu/libc.so.6
73024abb1000-73024ac00000 r--p 001b1000 08:30 46312 /usr/lib/x86_64-linux-gnu/libc.so.6
73024ac00000-73024ac04000 r--p 001ff000 08:30 46312 /usr/lib/x86_64-linux-gnu/libc.so.6
73024ac04000-73024ac06000 rw-p 00203000 08:30 46312 /usr/lib/x86_64-linux-gnu/libc.so.6
73024ac06000-73024ac13000 rw-p 00000000 00:00 0
73024ac77000-73024ac7a000 rw-p 00000000 00:00 0
73024ac82000-73024ac84000 rw-p 00000000 00:00 0
73024ac84000-73024ac88000 r--p 00000000 00:00 0 [vvar]
73024ac88000-73024ac8a000 r--p 00000000 00:00 0 [vvar_vclock]
73024ac8a000-73024ac8c000 r-xp 00000000 00:00 0 [vdso]
73024ac8c000-73024ac8d000 r--p 00000000 08:30 46309 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
73024ac8d000-73024acb0000 r-xp 00001000 08:30 46309 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
73024acb0000-73024acc2000 r--p 0002c000 08:30 46309 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
73024acc2000-73024acc4000 r--p 00036000 08:30 46309 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
73024acc4000-73024acc6000 rw-p 00038000 08:30 46309 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffc0d9ee000-7ffc0d9fb000 rw-p 00000000 00:00 0 [stack]
hari@LAPTOP-C63DNL02:~/OS/2520090048_Practical/Practical_7$
```